

8.04 Further Vectors			
8.04a	Vector product	a) Understand and be able to use the definition, in geometrical terms, of the vector product and be able to form the vector product in magnitude and direction, and in component form. <i>Includes use of the formula $\mathbf{a} \times \mathbf{b} = \mathbf{a} \mathbf{b} \sin \theta \hat{\mathbf{n}}$, where $\mathbf{a}$, $\mathbf{b}$, $\hat{\mathbf{n}}$, in that order (and the vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, in that order) form a right-handed triple.</i>	
8.04b		b) Understand the anti-commutative and distributive properties of the vector product.	
8.04c		c) Be able to use the vector product to calculate areas of triangles and parallelograms.	
8.04e			e) Understand and be able to use the definition of scalar triple product, and be able to use it to calculate volumes of tetrahedra and parallelepipeds.
8.04d		d) Understand the significance of $\mathbf{a} \times \mathbf{b} = \mathbf{0}$. <i>e.g. The equation of a line in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{d} = \mathbf{0}$.</i>	<i>Includes the notation $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.</i> <i>Includes understanding the significance of $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) =$</i>